W^EI sets the cliff; W^IE shapes the basin

nb036 · 30 May 2026 · revising · Gamma-gated sparsity

The trained networks this entry uses are produced once in the shared training hub, nb022 (Training), and reused here rather than retrained.

Abstract

Maps the (wEI, wIE) coupling plane: which weight controls the recruitment cliff, and where do *trained* networks land when initialised across it? *wEI* moves the cliff (E→I drive is the recruitment knob); *wIE* shapes the basin above engagement but does not move the cliff. Training from each grid point lands in two distinct basins — a low-E PING corner and a high-E silent-I stretched-COBA corner.

Method

Parameter	Value
Integration timestep Δt	0.1 ms
Trial duration T	200 ms
MNIST samples (80/20 stratified split of 2000)	1600 train / 400 test ($\approx$ 2.9% of the 70k-sample MNIST corpus)
Epochs	10

PING and COBA baseline definitions, training recipe, and the spike-budget regulariser are in nb025. The recruitment cliff at f^* is introduced in nb025. This entry asks: which part of the E↔I coupling architecture controls the cliff's position, and what happens when we train the network from coupling initialisations spanning both sides of it? Three experiments.

Coupling sweep (inference-time). If f^* is set by the E↔I coupling, altering W^{EI} or W^{IE} at inference should move it. On the trained PING baseline (θ_u = off), multiply either W^{EI} or W^{IE} by a scalar in $\{0.25, 0.5, 1.0, 2.0\}$ and re-run the W_{in} scale sweep on top. All other weights frozen.

Training grid. A stronger test is to train one network per grid point with W^{EI} and W^{IE} initialised to specified values from scratch, under heavy spike penalty: do the resulting solutions cluster into a PING-like region and a COBA-like region? 5×5 grid over $W^{EI}, W^{IE} \in \{0, 0.25, 0.5, 1.0, 2.0\}$ (mean initialisation, std fixed at $0.1 \times$ mean). One PING-architecture training run per cell at $\theta_u = 0.2$ from epoch 0, $W_{\text{in}} = 0.6$ (a compromise between COBA's 0.3 init and PING's 1.2 — gives the no-loop cells a fair shot at converging), **100 training epochs** per cell, seed 42. All other hyperparameters from the standard PING recipe. 25 trainings on Modal A100 in parallel. Reporting per-cell best-epoch accuracy.

Diagonal slice. The grid varies W^{EI} and W^{IE} independently, but the codebase's standard PING recipe ties them via $W^{IE} = \text{ei\_ratio} \cdot W^{EI}$ with $\text{ei\_ratio} = 2$. Restricting to that line gives a finer-grained view: one knob (W^{EI}) at 10 values $\{0, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 1.0, 1.5, 2.0\}$, W^{IE} tracking at twice the value, **three seeds (42, 43, 44) per value, 100 training epochs**. Best-epoch accuracy reported. Same recipe as the grid otherwise ($\theta_u = 0.2$, $W_{\text{in}} = 0.6$).

Results

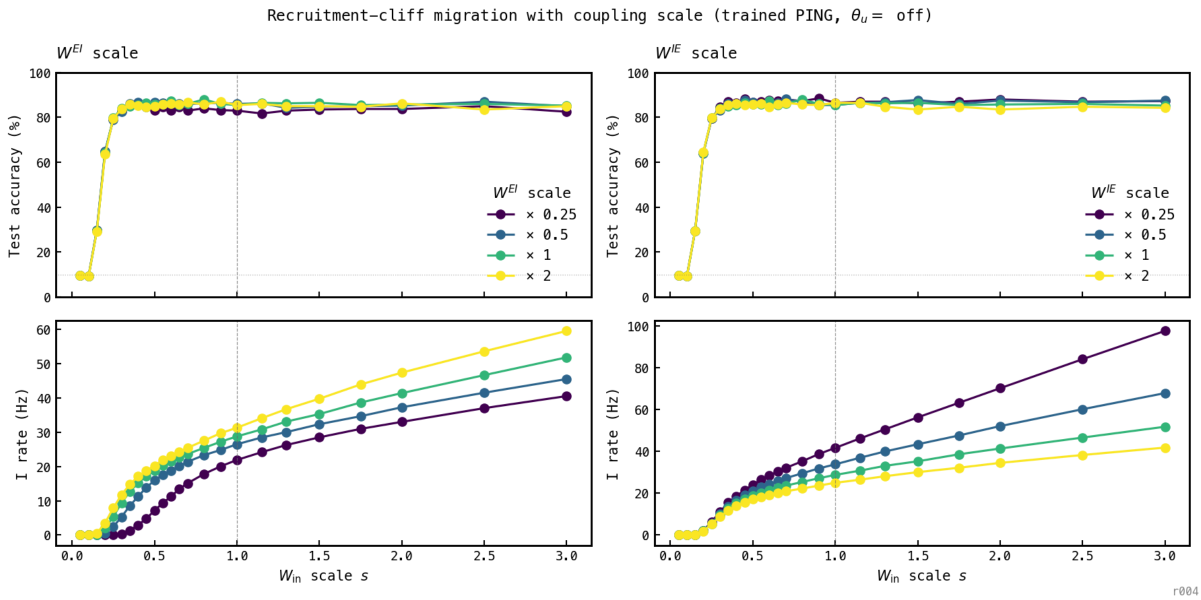

Figure 94: Inference-time W_{in} scale sweep on trained PING ($\theta_u =$ off) with either W^{EI} or W^{IE} scaled. Top: accuracy. Bottom: I rate. Left: W^{EI} sweep, W^{IE} fixed. Right: W^{IE} sweep, W^{EI} fixed.

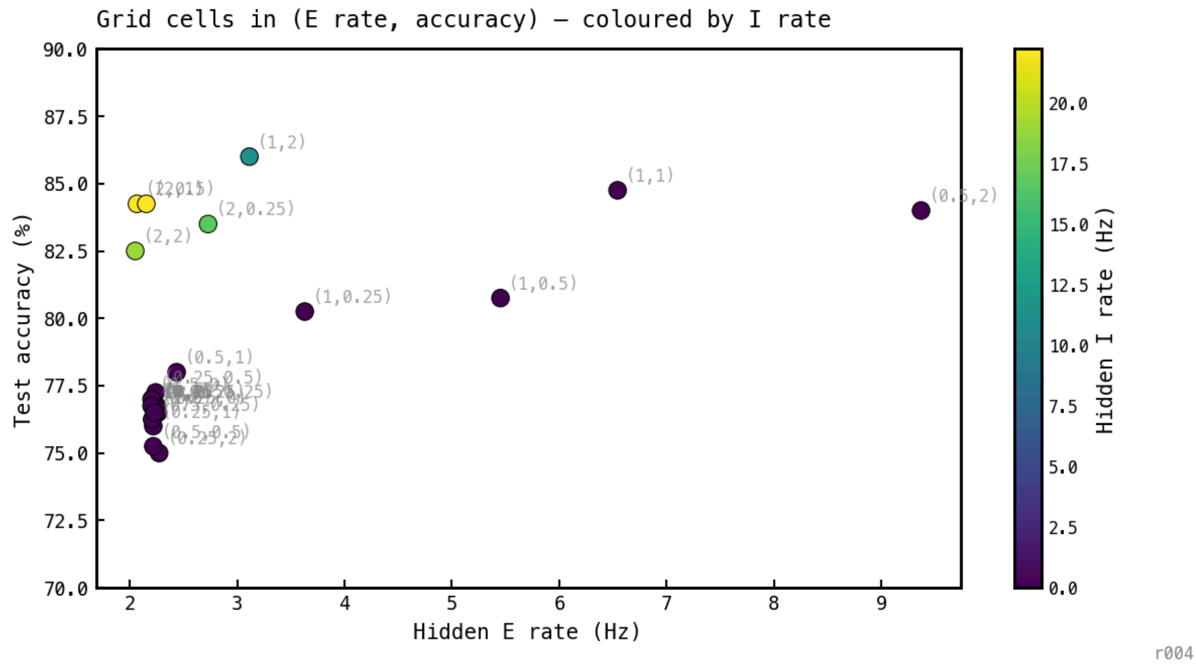

Figure 95: Each dot is one cell of the 5×5 (W^{EI}, W^{IE}) training grid, plotted at its best-epoch test accuracy vs final hidden-E rate. Colour encodes the third quantity — hidden-I rate — which is what discriminates the clusters. Labels are (W^{EI}, W^{IE}).

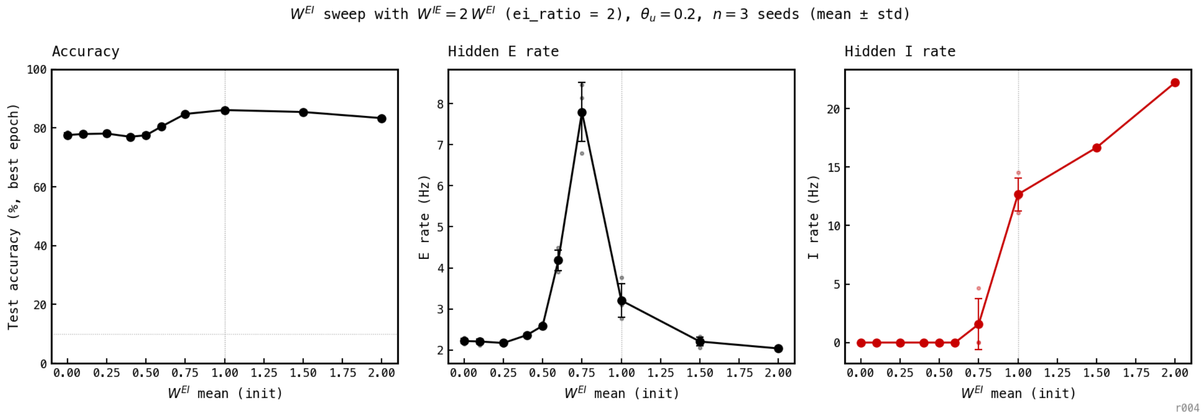

Figure 96: W^{EI} scanned over $\{0, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 1.0, 1.5, 2.0\}$ with $W^{IE} = 2W^{EI}$. Error bars are mean $\pm$ std of the best-epoch test accuracy over 3 seeds (42, 43, 44), 100 training epochs each; faint dots show individual seed peaks. **Left:** test accuracy. **Middle:** hidden-E rate (last-epoch). **Right:** hidden-I rate (last-epoch). Grey dotted vertical at $W^{EI} = 1$ marks the canonical recipe baseline.